Pandelis Zembashis

CTO | Staff Infrastructure Engineer | Cluster, Runtime & AI Platform Systems

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EXECUTIVE SUMMARY

CTO and staff-level infrastructure engineer building cluster-backed AI systems, secure execution platforms, and developer infrastructure. At Cosine, lead product and engineering for an applied AI lab whose agents run across local machines, Kubernetes and gVisor remote sandboxes, private cloud, on-premise, and air-gapped deployments. Strong fit for Anthropic Cluster Infrastructure: compute lifecycle strategy, secure cluster substrates, operational excellence, private deployment, reliability, and mentoring engineers through ambiguous multi-system work.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – Present

Applied AI lab building autonomous coding agents, secure execution infrastructure, and enterprise deployments.

- Own product and engineering for Cosine's AI coding platform, setting technical direction across runtime, cloud agents, deployment architecture, security boundaries, reliability, and product needs.
- Built the cluster-backed remote execution path for long-running coding agents: Kubernetes-scheduled workers, gVisor/runsc isolation, ephemeral per-session environments, WebSocket control planes, and secure shell/tool execution.
- Designed lifecycle and recovery patterns for local-to-remote sessions, including checkpointing, deterministic replay, event logs, task state propagation, and debugging hooks for failure analysis.
- Shaped secure-by-default deployment models across managed cloud, private VPC, on-prem, and fully air-gapped environments where code, logs, model artifacts, and tenant data must remain isolated.
- Led infrastructure decisions spanning network boundaries, tenant isolation, RBAC/SSO, auditability, image/runtime footprint, cold starts, cost controls, and operational support for regulated customers.
- Built 'cos2' as a single Go harness shared by CLI, desktop, VS Code/WebAssembly, API-server sessions, and remote worker pods, reducing cross-surface drift and making reliability fixes flow through one release path.
- Mentored engineers and aligned product, enterprise, ML, and infrastructure work around a small number of high-leverage platform bets.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled outbound email and event-driven automation systems that processed millions of email chains and billions of tokens.
- Established engineering cadence, incident response habits, onboarding, and operating rhythms during rapid seed-to-Series A growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a fleet management platform for light electric vehicles in Next.js, React Query, and Tailwind CSS.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving product iteration across web, operations, and marketplace workflows.
- Established development cadences, product intake, and stakeholder feedback loops that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021

White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

Substrakt | Software Developer | Aug. 2018 – Jun. 2019
Digital product agency building bespoke publishing, arts, and cultural web platforms.

- Built bespoke WordPress features, search experiences, and internal tooling for design-led client websites.
- Modernized the JavaScript toolchain from Gulp to Webpack, improving development and deployment workflows.

D4 Software | Software Engineer | Jun. 2017 – Jul. 2018
Product engineering consultancy building client platforms, tools, and cloud infrastructure.

- Built client-facing features and internal tools across product and platform engagements using JavaScript and .NET.
- Maintained AWS infrastructure, serverless services, databases, and virtual machines while modernizing the frontend stack with Webpack, TypeScript, and Vue.js.

WRITING & PUBLIC WORK

Why we built our own harness: Technical deep dive on Cosine's Go runtime, gVisor-sandboxed Kubernetes substrate, tool execution, replay, multi-agent orchestration, and ML feedback loop.
Cosine CLI runtime: Single-binary distribution, native search, filesystem performance, and reliable developer-tool execution.
Tech.eu on Cosine: Coverage of Cosine's full-stack AI product strategy, benchmark-leading models, and sovereign deployment work.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

Cluster & Runtime Infrastructure: Go, Kubernetes, gVisor/runsc, containers, remote workers, lifecycle automation, WebSockets, protobuf, Linux systems
Cloud, Network & Security: AWS, VPC/on-prem/air-gapped deployment, tenant isolation, RBAC/SSO, audit logs, network policy, encryption, secure execution
Reliability & Operations: Observability, incident response, postmortem culture, checkpointing, deterministic replay, cost/cold-start optimization, failure analysis
Technical Leadership: Staff-level strategy, roadmaps, stakeholder alignment, mentorship, hiring, cross-functional infrastructure tradeoffs